Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Vierra Dairy Farms #3

Physical address of dairy:

Number and Street	City	County	Zip Code
7884 Faith Home RD	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X190-X001-XXXX

B. OPERATORS

Swanson, Matt

Operator name: Swanson, Matt

Telephone no.: (209) 668-0343 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1350	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Ferreira, Eleanor

Legal owner name: Ferreira, Eleanor

Telephone no.: (209) 632-1538 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
1781 N Johnson RD	Turlock	CA	95380

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	262	0	0
Number under roof	304	0	0	0	0	0
Maximum number	304	0	0	262	0	0
Average number	304	0	0	262	0	0
Avg live weight (lbs)	1,400	0	0	700		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 10,350.48 *tons per reporting period*

Total nitrogen from manure: 127,752.30 *lbs per reporting period*

After ammonia losses (30% loss applied): 89,426.61 *lbs per reporting period*

Total phosphorus from manure: 21,639.73 *lbs per reporting period*

Total potassium from manure: 58,618.26 *lbs per reporting period*

Total salt from manure: 143,138.40 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,916,000 *gallons*

Total nitrogen generated: 27,595.18 *lbs*

Total phosphorus generated: 4,428.52 *lbs*

Total potassium generated: 34,052.61 *lbs*

Total salt generated: 247,313.76 *lbs*

	8,916,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,916,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/09/2023	Solid commercial fertilizer *UN32*	1.80 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	1,150.85	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	1,150.85	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
B10	20	20	2	process wastewater	X045-X040-X036-XXXX
B11	20	20	2	process wastewater	X045-X040-X036-XXXX
B6	3	3	0	none	X045-X190-X001-XXXX
B7	17	17	2	process wastewater	X045-X190-X001-XXXX
B8	2	2	1	manure	X045-X190-X001-XXXX
B9	18	18	2	manure	X045-X190-X009-XXXX
Totals for areas that were used for application	77	77	9		
Totals for areas that were not used for application	3	3	0		
Land application area totals	80	80	9		

B. CROPS AND HARVESTS

B10

Field name: B10

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	320.40 *ton*	As-is		75.0	4,400.00	800.00	5,000.00		9.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.02	140.98	25.63	160.20	758.55

B10

05/27/2023: Corn, silage

Crop: Corn, silage — Acres planted: 20 — Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	593.60 *ton*	As-is		70.4	4,600.00	800.00	5,100.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.68	273.06	47.49	302.74	1,001.52

B11

Field name: B11

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 20 — Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	281.60 *ton*	As-is		77.0	4,900.00	900.00	6,400.00		9.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.08	137.98	25.34	180.22	586.15

05/27/2023: Corn, silage

Crop: Corn, silage — Acres planted: 20 — Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	667.20 *ton*	As-is		70.4	3,600.00	700.00	4,600.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.36	240.19	46.70	306.91	2,014.41

B7

Field name: B7

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 17 — Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	218.40 *ton*	As-is		71.8	5,200.00	700.00	4,400.00		7.72

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.85	133.61	17.99	113.05	559.37

06/20/2023: Corn, silage

Crop: Corn, silage — Acres planted: 17 — Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/30/2023	360.57 *ton*	As-is		70.4	4,600.00	800.00	5,100.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.21	195.13	33.94	216.34	715.71

B8

Field name: B8

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 2 — Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	23.84 *ton*	As-is		63.8	4,800.00	1,000.00	5,200.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.92	114.43	23.84	123.97	681.78

B9

Field name: B9

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	231.66 *ton*	As-is		62.6	4,600.00	1,000.00	5,300.00		8.77

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.87	118.40	25.74	136.42	844.27

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	574.02 *ton*	As-is		71.8	3,800.00	700.00	4,000.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.89	242.36	44.65	255.12	1,546.79

NUTRIENT BUDGET

A. LAND APPLICATIONS

B10 - 11/15/2022: Oats, silage-soft dough

Field name: B10

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.62	4.98	70.59	475.67	228,000.00 *gal*
Application event totals		42.62	4.98	70.59	475.67	

01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.79	6.79	50.11	258.36	144,000.00 *gal*
Application event totals		51.79	6.79	50.11	258.36	

02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.84	5.09	37.58	193.77	108,000.00 *gal*
Application event totals		38.84	5.09	37.58	193.77	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.53	4.53	33.41	172.24	96,000.00 *gal*
Application event totals		34.53	4.53	33.41	172.24	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.83	1.13	37.60	402.56	120,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	286.11	3,516,440.00 *gal*
Application event totals		24.83	1.13	37.60	688.67	

B10 - 05/27/2023: Corn, silage

Field name: B10

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.27	4.96	165.45	1,771.28	528,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	382.63	4,702,737.00 *gal*
Application event totals		109.27	4.96	165.45	2,153.91	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	361.55	4,443,646.00 *gal*
Application event totals		0.00	0.00	0.00	361.55	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.69	15.33	63.79	338.57	588,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	182.08	2,237,816.00 *gal*
Application event totals		55.69	15.33	63.79	520.65	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.64	10.64	44.26	234.93	408,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	134.35	1,651,196.00 *gal*
Application event totals		38.64	10.64	44.26	369.28	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.46	12.52	52.07	276.39	480,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	184.95	2,273,152.00 *gal*
Application event totals		45.46	12.52	52.07	461.34	

B10 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	168.11	2,066,206.00 *gal*
Application event totals		0.00	0.00	0.00	168.11	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.87	13.46	55.98	297.12	516,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	120.43	1,480,099.00 *gal*
Application event totals		48.87	13.46	55.98	417.54	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	137.26	1,687,045.00 *gal*
Application event totals		0.00	0.00	0.00	137.26	

B11 - 11/15/2022: Oats, silage-soft dough

Field name: B11

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.42	7.92	58.46	301.42	168,000.00 *gal*
Application event totals		60.42	7.92	58.46	301.42	

01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.48	6.22	45.93	236.83	132,000.00 *gal*
Application event totals		47.48	6.22	45.93	236.83	

B11 - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.21	3.96	29.23	150.71	84,000.00 *gal*
Application event totals		30.21	3.96	29.23	150.71	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.53	4.53	33.41	172.24	96,000.00 *gal*
Application event totals		34.53	4.53	33.41	172.24	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.90	0.68	22.56	241.54	72,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	286.11	3,516,440.00 *gal*
Application event totals		14.90	0.68	22.56	527.65	

B11 - 05/27/2023: Corn, silage

Field name: B11

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	131.62	5.97	199.29	2,133.58	636,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	382.63	4,702,737.00 *gal*
Application event totals		131.62	5.97	199.29	2,516.22	

B11 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	361.55	4,443,646.00 *gal*
Application event totals		0.00	0.00	0.00	361.55	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.74	13.14	54.68	290.21	504,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	182.08	2,237,816.00 *gal*
Application event totals		47.74	13.14	54.68	472.28	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.92	11.27	46.87	248.75	432,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	134.35	1,651,196.00 *gal*
Application event totals		40.92	11.27	46.87	383.10	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.64	10.64	44.26	234.93	408,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	184.95	2,273,152.00 *gal*
Application event totals		38.64	10.64	44.26	419.88	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	168.11	2,066,206.00 *gal*
Application event totals		0.00	0.00	0.00	168.11	

B11 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.19	11.89	49.47	262.57	456,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	120.43	1,480,099.00 *gal*
Application event totals		43.19	11.89	49.47	382.99	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	137.26	1,687,045.00 *gal*
Application event totals		0.00	0.00	0.00	137.26	

B7 - 10/10/2022: Oats, silage-soft dough

Field name: B7

Crop: Oats, silage-soft dough Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.95	4.31	61.19	412.34	168,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	253.57	2,648,974.00 *gal*
Application event totals		36.95	4.31	61.19	665.91	

12/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.11	2.46	34.97	235.62	96,000.00 *gal*
Application event totals		21.11	2.46	34.97	235.62	

B7 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.54	4.66	34.39	177.31	84,000.00 *gal*
Application event totals		35.54	4.66	34.39	177.31	

02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.62	5.33	39.30	202.64	96,000.00 *gal*
Application event totals		40.62	5.33	39.30	202.64	

03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.31	2.66	19.65	101.32	48,000.00 *gal*
Application event totals		20.31	2.66	19.65	101.32	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.22	1.33	44.24	473.60	120,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	289.81	3,027,611.00 *gal*
Application event totals		29.22	1.33	44.24	763.41	

B7 - 06/20/2023: Corn, silage

Field name: B7

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

B7 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.42	4.37	145.99	1,562.89	396,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	323.14	3,375,811.00 *gal*
Application event totals		96.42	4.37	145.99	1,886.03	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	273.54	2,857,630.00 *gal*
Application event totals		0.00	0.00	0.00	273.54	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.84	16.20	67.39	357.68	528,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	214.27	2,238,420.00 *gal*
Application event totals		58.84	16.20	67.39	571.94	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.78	10.68	44.42	235.74	348,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	231.74	2,420,924.00 *gal*
Application event totals		38.78	10.68	44.42	467.48	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	175.47	1,833,103.00 *gal*
Application event totals		0.00	0.00	0.00	175.47	

B7 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.13	12.15	50.54	268.26	396,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	175.47	1,833,103.00 *gal*
Application event totals		44.13	12.15	50.54	443.73	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.14	13.25	55.14	292.64	432,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	165.60	1,730,016.00 *gal*
Application event totals		48.14	13.25	55.14	458.25	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	181.40	1,895,024.00 *gal*
Application event totals		0.00	0.00	0.00	181.40	

B8 - 10/10/2022: Oats, silage-soft dough

Field name: B8

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	141.84	33.84	53.28	2,628.00	32.00 *ton*
Application event totals		141.84	33.84	53.28	2,628.00	

B8 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	305.22	375,132.00 *gal*
Application event totals		0.00	0.00	0.00	305.22	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	223.89	275,176.00 *gal*
Application event totals		0.00	0.00	0.00	223.89	

B9 - 10/10/2022: Oats, silage-soft dough

Field name: B9

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	141.84	33.84	53.28	2,628.00	288.00 *ton*
Application event totals		141.84	33.84	53.28	2,628.00	

09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	297.07	3,286,008.00 *gal*
Application event totals		0.00	0.00	0.00	297.07	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	252.20	2,789,704.00 *gal*
Application event totals		0.00	0.00	0.00	252.20	

B9 - 05/30/2023: Corn, silage

Field name: B9

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	210.59	66.55	61.59	1,619.13	600.00 *ton*
Application event totals		210.59	66.55	61.59	1,619.13	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	292.61	3,236,700.00 *gal*
Application event totals		0.00	0.00	0.00	292.61	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	270.22	2,989,016.00 *gal*
Application event totals		0.00	0.00	0.00	270.22	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	240.42	2,659,344.00 *gal*
Application event totals		32.00	0.00	0.00	240.42	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	182.13	2,014,575.00 *gal*
Application event totals		32.00	0.00	0.00	182.13	

B9 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	206.29	2,281,813.00 *gal*
Application event totals		0.00	0.00	0.00	206.29	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	183.60	2,030,870.00 *gal*
Application event totals		0.00	0.00	0.00	183.60	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	112.89	1,248,710.00 *gal*
Application event totals		0.00	0.00	0.00	112.89	

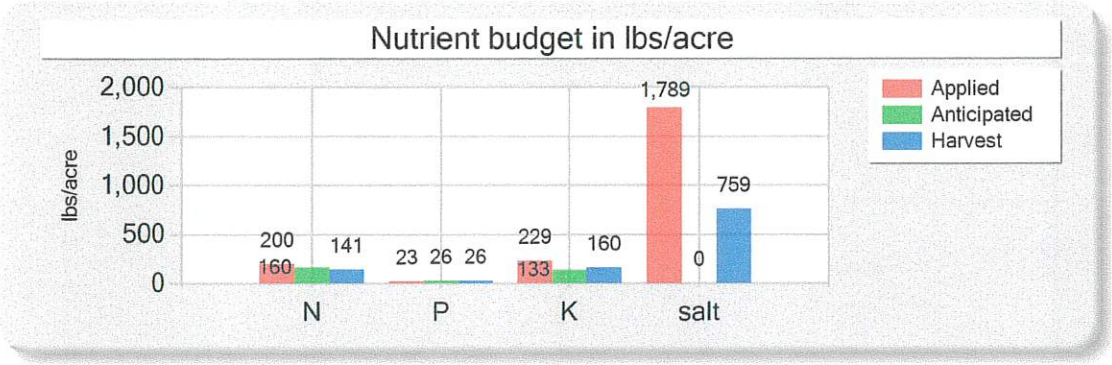
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	101.10	1,118,350.00 *gal*
Application event totals		0.00	0.00	0.00	101.10	

B. NUTRIENT BUDGET

B10 - 11/15/2022: Oats, silage-soft dough

Field name: B10 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.62	22.51	229.29	1,502.60
Fresh water	0.00	0.00	0.00	286.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.62	22.51	229.29	1,788.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	140.98	25.63	160.20	758.55
Nutrient balance	58.64	-3.12	69.09	1,030.16
Applied to removed ratio	1.42	0.88	1.43	2.36

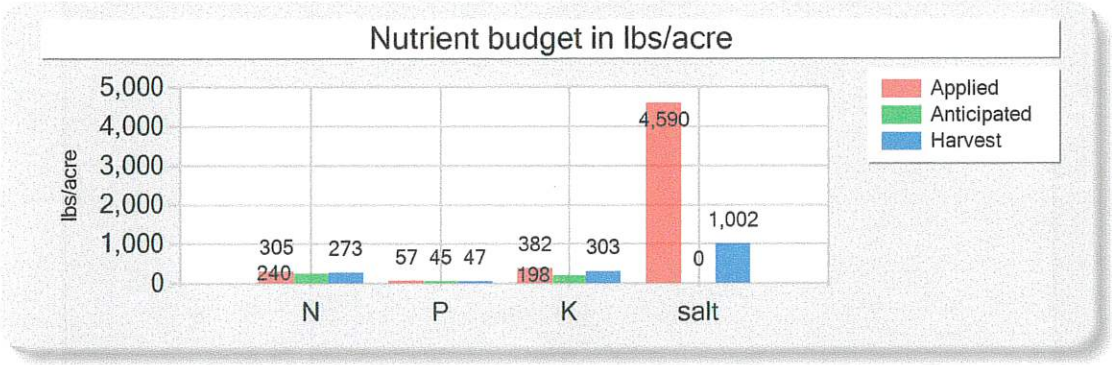
Fresh water applied
3,516,440.00 *gallons*
129.50 *acre-inches*
6.47 *inches/acre*

Process wastewater applied
696,000.00 *gallons*
25.63 *acre-inches*
1.28 *inches/acre*

Total harvests for the crop
1 *harvests*

B10 - 05/27/2023: Corn, silage

Field name: B10 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	297.95	56.90	381.55	2,918.28
Fresh water	0.00	0.00	0.00	1,671.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.95	56.90	381.55	4,589.65
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	273.06	47.49	302.74	1,001.52
Nutrient balance	31.89	9.42	78.82	3,588.12
Applied to removed ratio	1.12	1.20	1.26	4.58

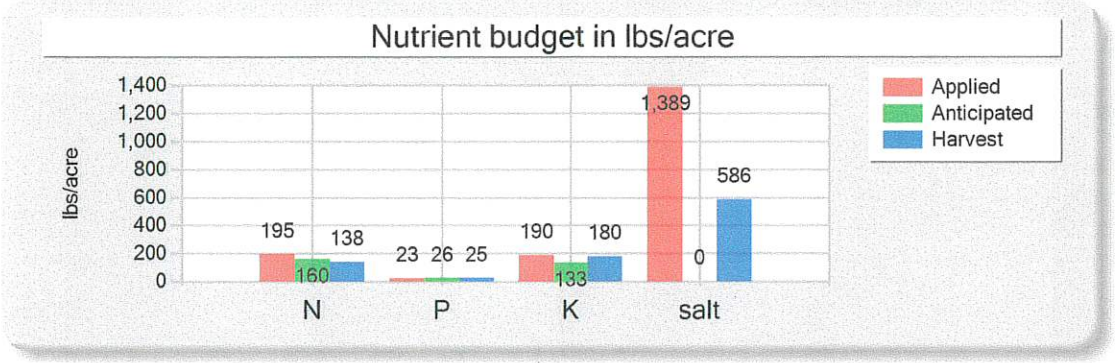
Fresh water applied
20,541,897.00 *gallons*
756.49 *acre-inches*
37.82 *inches/acre*

Process wastewater applied
2,520,000.00 *gallons*
92.80 *acre-inches*
4.64 *inches/acre*

Total harvests for the crop
1 *harvests*

B11 - 11/15/2022: Oats, silage-soft dough

Field name: B11 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	187.54	23.31	189.60	1,102.74
Fresh water	0.00	0.00	0.00	286.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.54	23.31	189.60	1,388.85
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	137.98	25.34	180.22	586.15
Nutrient balance	56.56	-2.04	9.37	802.70
Applied to removed ratio	1.41	0.92	1.05	2.37

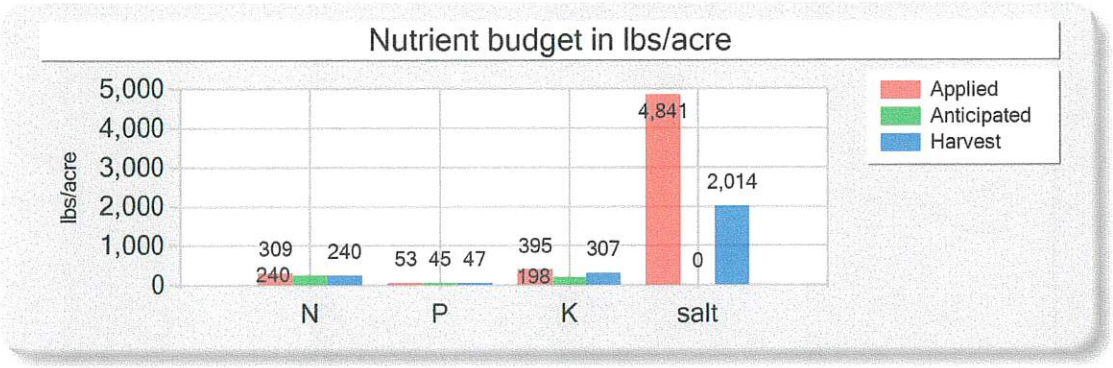
Fresh water applied
3,516,440.00 *gallons*
129.50 *acre-inches*
6.47 *inches/acre*

Process wastewater applied
552,000.00 *gallons*
20.33 *acre-inches*
1.02 *inches/acre*

Total harvests for the crop
1 *harvests*

B11 - 05/27/2023: Corn, silage

Field name: B11 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	302.11	52.91	394.57	3,170.03
Fresh water	0.00	0.00	0.00	1,671.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	309.11	52.91	394.57	4,841.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	240.19	46.70	306.91	2,014.41
Nutrient balance	68.92	6.21	87.65	2,826.99
Applied to removed ratio	1.29	1.13	1.29	2.40

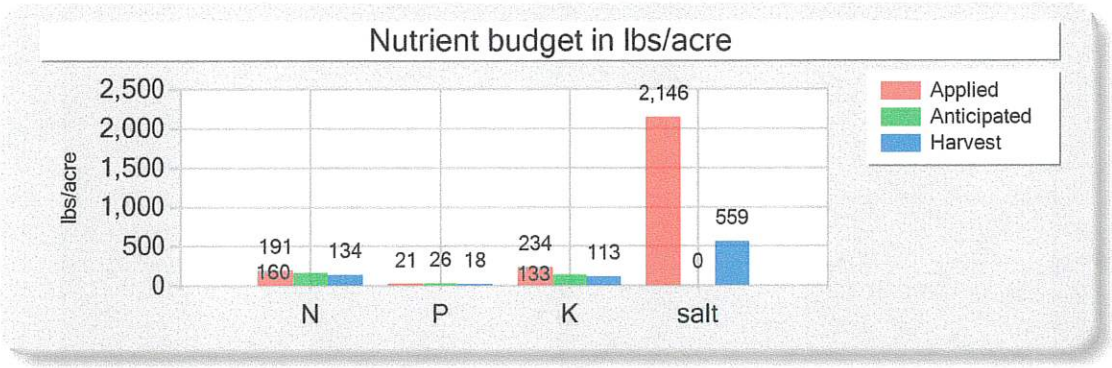
Fresh water applied
20,541,897.00 *gallons*
756.49 *acre-inches*
37.82 *inches/acre*

Process wastewater applied
2,436,000.00 *gallons*
89.71 *acre-inches*
4.49 *inches/acre*

Total harvests for the crop
1 *harvests*

B7 - 10/10/2022: Oats, silage-soft dough

Field name: B7 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.75	20.75	233.74	1,602.83
Fresh water	0.00	0.00	0.00	543.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.75	20.75	233.74	2,146.20
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	133.61	17.99	113.05	559.37
Nutrient balance	57.14	2.76	120.68	1,586.83
Applied to removed ratio	1.43	1.15	2.07	3.84

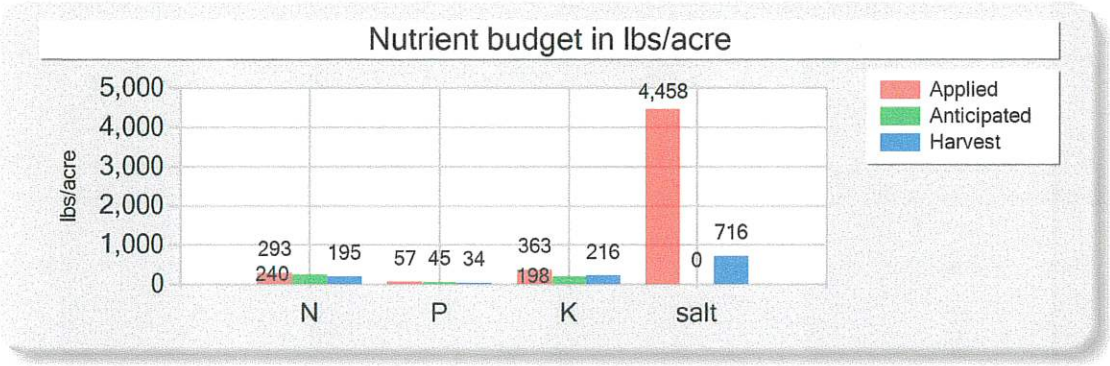
Fresh water applied
5,676,585.00 *gallons*
209.05 *acre-inches*
12.30 *inches/acre*

Process wastewater applied
612,000.00 *gallons*
22.54 *acre-inches*
1.33 *inches/acre*

Total harvests for the crop
1 *harvests*

B7 - 06/20/2023: Corn, silage

Field name: B7 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.29	56.65	363.47	2,717.21
Fresh water	0.00	0.00	0.00	1,740.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.29	56.65	363.47	4,457.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	195.13	33.94	216.34	715.71
Nutrient balance	98.16	22.72	147.12	3,742.11
Applied to removed ratio	1.50	1.67	1.68	6.23

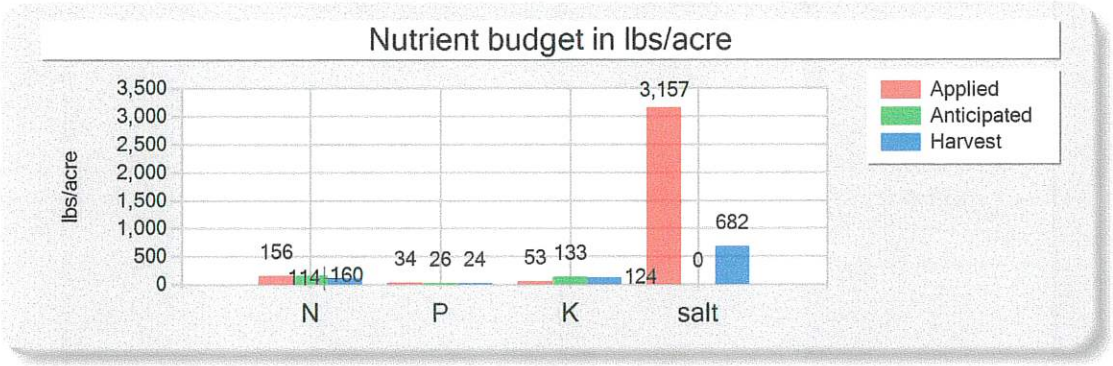
Fresh water applied
18,184,031.00 *gallons*
669.66 *acre-inches*
39.39 *inches/acre*

Process wastewater applied
2,100,000.00 *gallons*
77.34 *acre-inches*
4.55 *inches/acre*

Total harvests for the crop
1 *harvests*

B8 - 10/10/2022: Oats, silage-soft dough

Field name: B8 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.84	33.84	53.28	2,628.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	529.11
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	155.84	33.84	53.28	3,157.11
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	114.43	23.84	123.97	681.78
Nutrient balance	41.41	10.00	-70.69	2,475.34
Applied to removed ratio	1.36	1.42	0.43	4.63

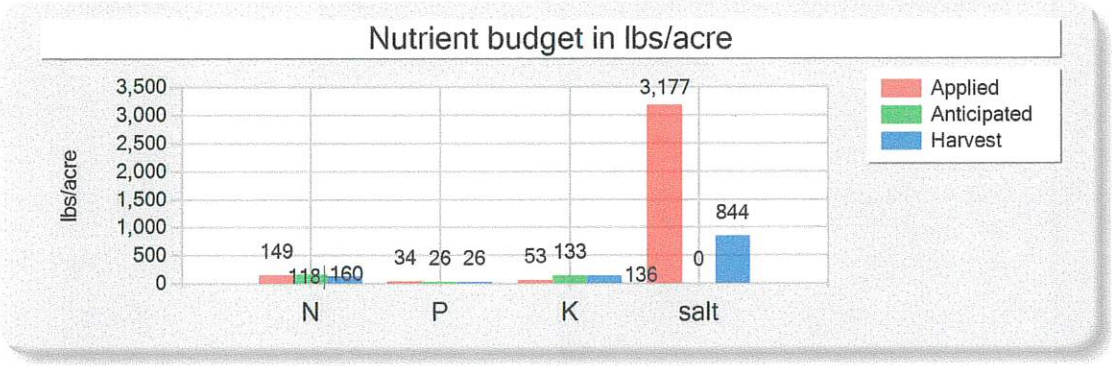
Fresh water applied
650,308.00 *gallons*
23.95 *acre-inches*
11.97 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

B9 - 10/10/2022: Oats, silage-soft dough

Field name: B9 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.84	33.84	53.28	2,628.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	549.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	148.84	33.84	53.28	3,177.27
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	118.40	25.74	136.42	844.27
Nutrient balance	30.44	8.10	-83.14	2,333.00
Applied to removed ratio	1.26	1.31	0.39	3.76

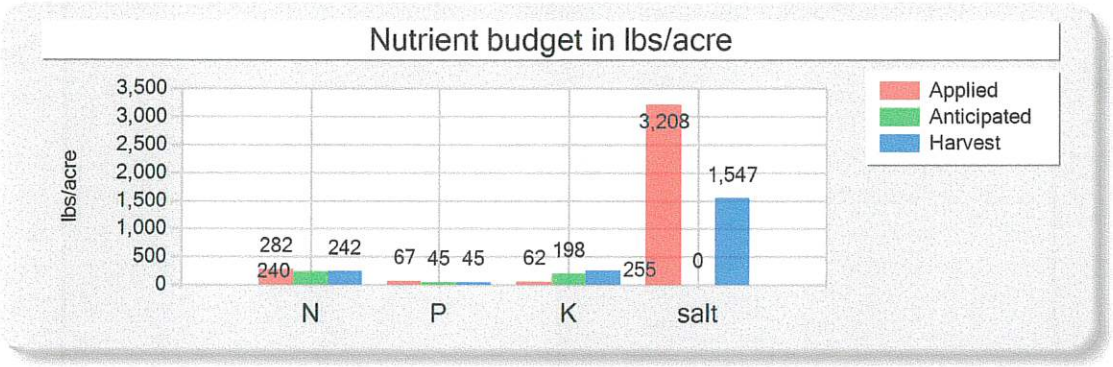
Fresh water applied
6,075,712.00 *gallons*
223.75 *acre-inches*
12.43 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

B9 - 05/30/2023: Corn, silage

Field name: B9 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	210.59	66.55	61.59	1,619.13
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,589.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.59	66.55	61.59	3,208.38
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	242.36	44.65	255.12	1,546.79
Nutrient balance	39.22	21.91	-193.53	1,661.59
Applied to removed ratio	1.16	1.49	0.24	2.07

Fresh water applied
17,579,378.00 *gallons*
647.39 *acre-inches*
35.97 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Separated Solids

Sample and source description: 2022 Fall Separated Solids

Sample date: 11/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	4,700.00	7,400.00	13,000.00	4,600.00	2,300.00	3,100.00	880.00		36.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 85.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	6,700.00	6,200.00							16.30
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,600.00	4,300.00	13,700.00	3,300.00	2,000.00	3,300.00	897.00		10.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	448.00	246.00	0.00	0.00	52.30	742.00								6,230.00	5,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	862.00	370.00	0.00	0.00	113.00	834.00								7,080.00	4,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	496.00	347.00	0.00	0.00	22.50	751.00								6,730.00	8,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	227.00	120.00	0.00	0.00	62.50	260.00	73.00	49.60	64.40	1,080.00	0.00	150.00	104.00	2,620.00	1,380
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	616.00	437.00	0.00	0.00	18.40	787.00								7,770.00	6,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

T.I.D Canal

Lateral 7

Sample description: Lateral 7

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	195
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

B10

Soil

Sample and source description: Soil

Sample date: 03/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			98.00				
DL			4.00				

B11

B11

Soil

Sample and source description: Soil

Sample date: 01/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			14.00				
DL			4.00				

B7

Soil

Sample and source description: Soil

Sample date: 04/03/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.00				
DL			4.00				

B8

Soil

Sample and source description: Soil

Sample date: 04/03/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.00				
DL			4.00				

B8

Soil

Sample and source description: Soil

Sample date: 04/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			94.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

B10 - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	800.00	5,000.00		9.47
DL	100.00	100.00	100.00		0.01

B10 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	800.00	5,100.00		5.70
DL	100.00	100.00	100.00		0.01

B11 - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	900.00	6,400.00		9.05
DL	100.00	100.00	100.00		0.01

B11 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	700.00	4,600.00		10.20
DL	100.00	100.00	100.00		0.01

B7 - 10/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	700.00	4,400.00		7.72
DL	100.00	100.00	100.00		0.01

B7 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	800.00	5,100.00		5.70
DL	100.00	100.00	100.00		0.01

B8 - 10/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,000.00	5,200.00		7.90
DL	100.00	100.00	100.00		0.01

B9 - 10/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	1,000.00	5,300.00		8.77
DL	100.00	100.00	100.00		0.01

B9 - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	700.00	4,000.00		8.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

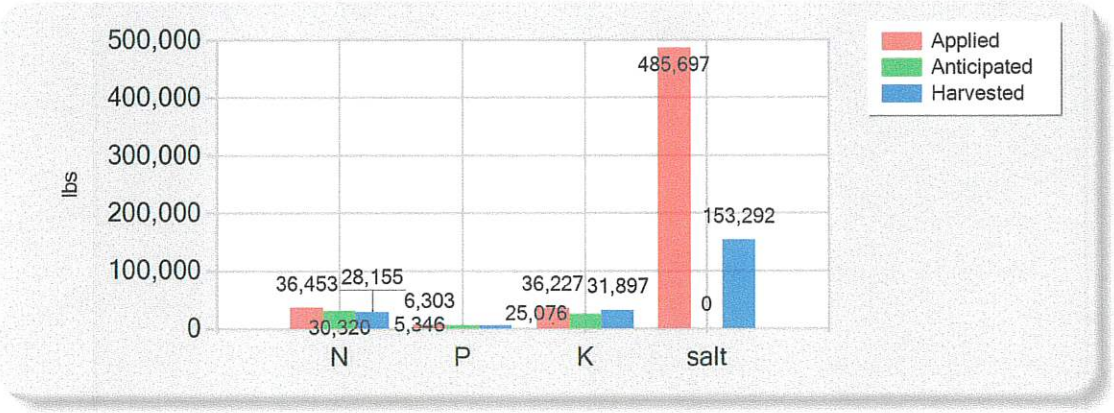
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

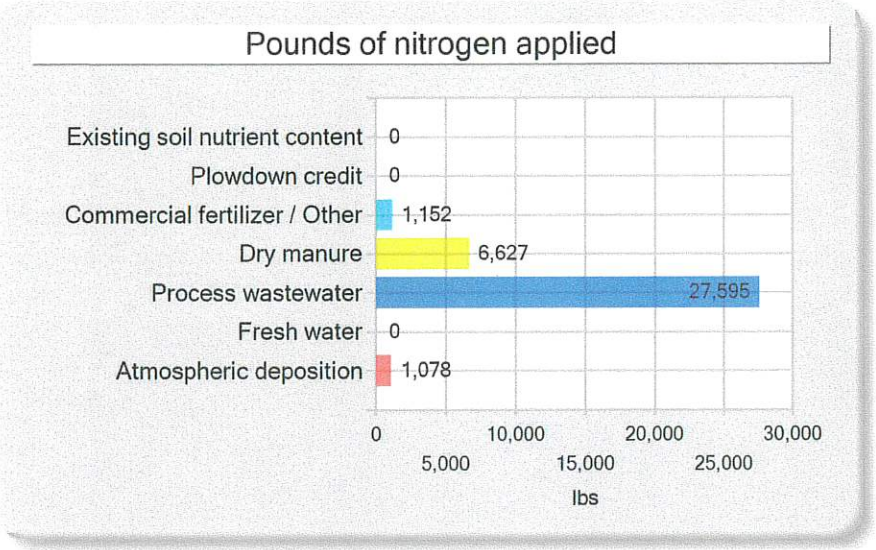
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

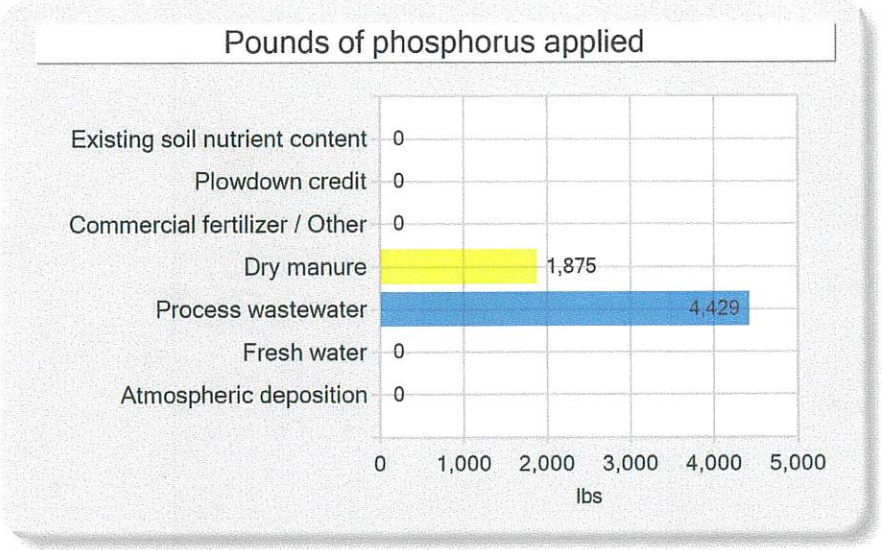
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,152.00	0.00	0.00	0.00
Dry manure	6,627.36	1,874.76	2,174.16	81,704.40
Process wastewater	27,595.18	4,428.52	34,052.61	247,313.76
Fresh water	0.00	0.00	0.00	156,678.41
Atmospheric deposition	1,078.00	0.00	0.00	0.00
Total nutrients applied	36,452.54	6,303.28	36,226.77	485,696.57
Anticipated crop nutrient removal	30,320.00	5,346.20	25,075.60	0.00
Actual crop nutrient removal	28,155.45	5,100.66	31,896.87	153,291.60
Nutrient balance	8,297.09	1,202.62	4,329.90	332,404.97
Applied to removed ratio	1.29	1.24	1.14	3.17

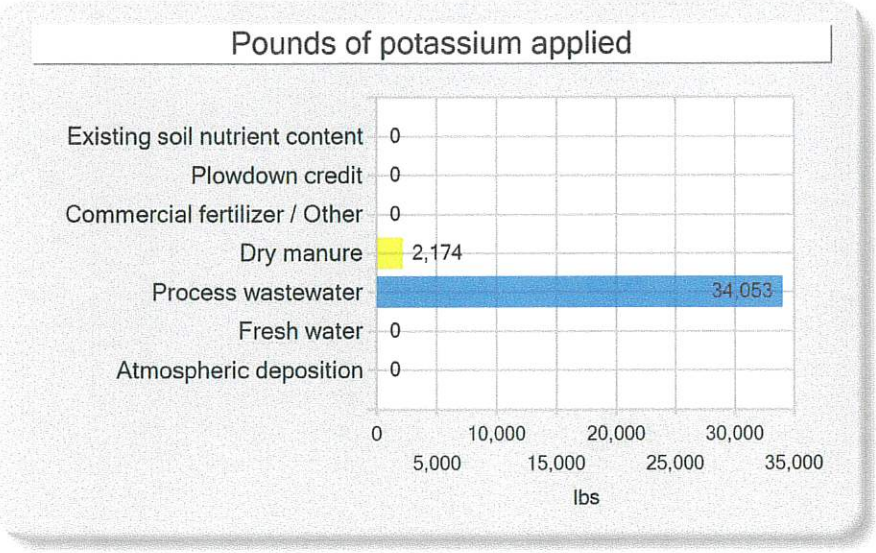
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

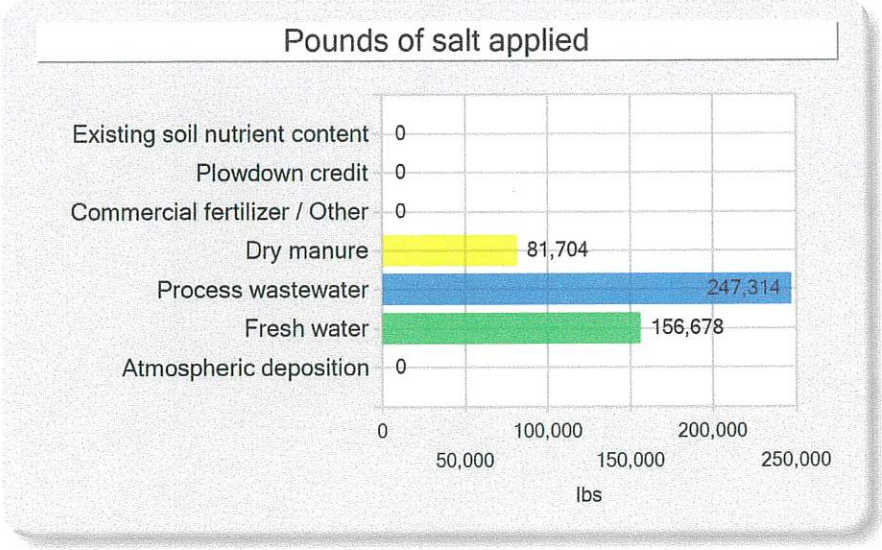


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field B7 Corn silage had lower than anticipated yields (21.2 tons per acre harvested vs 30 tons per acre anticipated) which resulted in this field exceeding the 1.4 ratio requirement.

Field B6 remained Fallow all year.
Field B8 was fallow during corn silage season.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Eleanore Ferreira

SIGNATURE OF OWNER OF FACILITY

Eleanor Ferreira

PRINT OR TYPE NAME

4/21/2024

DATE

SIGNATURE OF OPERATOR OF FACILITY

Matt Swanson

PRINT OR TYPE NAME

DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Vierra Dairy Farms #3
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FAR
Sampled By: JM
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

H1	H3	M3	F18	OTHER

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis				
1	12/31	11:45	Domestic	DW#1	16.4°C		V				
2			wells								
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Kathleen	FRAG				
2nd		DENELE ANALYTICAL, INC.	1-30-24 10:10			T4030008

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Vierra Dairy Farms #3
P.O. Box 1350
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/30/24 10:10 AM
SUBMITTED BY: KK
ANALYZED DATE: 1/31/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #1	T4030008A	Electrical Conductivity (EC)	EPA 120.1 and	1.31	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	839	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	103	101 (T4030004A)	98.8 (T4030004A)	1.80
Electrical Conductivity (EC)	99.1			0.959
Nitrate Nitrogen (NO3-N)	105	106 (T4030002A)	104 (T4030002A)	1.46

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